BTS–III–11.19 –1396 Reg. No. [ ]

A

B.Tech. Degree III Semester Examination November 2019

CE/CS/EC/EE/IT/ME/SE

AS 15-1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer ***ALL*** questions)

I. (a) By reducing to Echelon-form find the rank of the matrix

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

(b) Prove that every square matrix and its transpose have same eigen values.

(c) Show that the vectors (1,2,3), (3,-2,1), (1,-6,-5) are linearly dependent and find the relation connecting them.

(d) Explain basis and dimension of a vector space with example.

(e) Find the eigen values and eigen vectors of $A=\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

(f) Express $f(x)=|x|, -\pi < x < \pi$ as a Fourier series.

(g) Explain periodic function with two examples.

(h) Find $L\left[t\,e^{-t}\cosh ht\right]$.

(i) Find $L^{-1}\left[\log\dfrac{s(s+1)}{s^2+1}\right]$.

(j) Prove that $\beta(m,n)=\beta(n,m)$.

PART B

$(4 \times 10 = 40)$

II. (a) Test for consistency and hence solve
$x-2y+3z=2;\quad 2x-3z=3;\quad x+y+z=0$.

(b) State Cayley-Hamilton and verify it for the matrix $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

OR

III. (a) Diagonalize $A=\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ hence find A^6.

(b) Reduce the quadratic form $2xy+2yz+2zx$ into canonical form.

(P.T.O.)

IV. (a) Give an example of a vector space with dimension n. Justify your answer.

(b) Explain inner product space with an example.

OR

V. (a) Apply the Gram-Schmidt orthogonalization process to find an orthogonal basis for the subspace V of R^3 spanned by $V_1 = (0,1,2)$ $V_2 = (1,1,2)$ $V_3 = (1,0,1)$.

(b) Find k so that $u = (1, 2, k, 3)$ *and* $v = (3, k, 7, -5)$ in R^4 are orthogonal.

VI. (a) Obtain the half range sine series of the function $f(x) = kx(x-l)$ *in* $0 \le x \le l$.

(b) Using Fourier sine integral for $f(x) = e^{-ax} (a > o)$ show that

$$\int_0^\infty \frac{\lambda \sin \lambda x}{\lambda^2 + a^2} d\lambda = \frac{\pi}{2} e^{-ax}.$$

OR

VII. (a) Obtain a Fourier expansion for $\sqrt{1-\cos x}$ in the interval $-\pi < x < \pi$.

(b) Find the Fourier sine transform of $f(x) = \begin{cases} x & in\ \ 0 < x < 1 \\ 2-x & in\ \ 1 < x < 2 \\ 0 & in\ \ x > 2 \end{cases}$

VIII. (a) State convolution theorem and hence find $L^{-1}\left[\frac{s^2}{\left[s^2+a^2\right]\left[s^2+b^2\right]}\right]$.

(b) Solve (using Laplace transform) $y'' + 2y' - 3y = \sin t$ given $y = 0, y'(0) = 0$ when $t = 0$.

OR

IX. (a) Prove that $\beta(m,n) = \frac{\overline{|m}\,\overline{|n}}{\overline{|(m+n)}}$.

(b) Evaluate $\int_0^\infty \frac{e^{-2t} - e^{-3t}}{t} dt$.

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BTS-III(S)-04.19-0776 Reg. No. [| | | | | | |]

C

B.Tech. Degree III Semester Supplementary Examination April 2019

CE/CS/EC/EE/IT/ME/SE
AS 15-1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES
(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$

(b) Explain linear system of equations and when this system has unique solution and no solution.

(c) Explain linearly independent and dependent set of vectors in a vector space with example.

(d) State and prove one property of Eigen value of a square matrix

(e) Determine whether the following set of vectors form a basis in R^3 .

(f) Express $f(x) = |x|, -\pi \le x \le \pi$ as a Fourier series.

(g) Find the Fourier transform of $e^{-a|t|}$

(h) Find Laplace transform of $\frac{1-\cos 2t}{t}$

(i) Apply convolution theorem to evaluate $L^{-1}\frac{s}{(s^2+a^2)^2}$

(j) Find the inverse Laplace transform of $log\left(\frac{s+1}{s-1}\right)$

PART B

$(4 \times 10 = 40)$

II. (a) Define augmented matrix .Check whether the given system of equations is consistent or not.If consistent, solve it
$3x + y + z = 2, x - 3y + 2z = 1, 7x - y + 4z = 5$

(b) Define linear combination of vectors. If possible, write $V = (2, -5, 3)$ in R^3 as a linear combination of the vectors $e_1 = (1, -3, 2), e_2 = (2, -4, -1)$, $e_3 = (1, -5, 7)$

OR

III. (a) State Cayley Hamilton theorem. Using it, find the inverse of
$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

(b) Define linear transformation of a mapping T.Let $T: R^3 \to R^3$ defined by $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ z \\ x-y \end{bmatrix}$.Find Range of T, kernel of T , rank of T

(P.T.O.)

IV. (a) Find Eigen values and Eigen vectors of $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

(b) Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to the canonical form.

OR

V. (a) If x, y, z are linearly independent vectors of a vector space V .Prove that $x, x + y, x + y + z$ are also linearly independent.

(b) Find a basis and dimension of $M_{2\times2}$

VI. (a) Find a Fourier series to represent $x - x^2$ from $x = -\pi$ to $x = \pi$. Hence show that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}$

(b) Find Fourier cosine and sine transform of $f(x) = e^{-ax}$

OR

VII. (a) Using Fourier integral show that $\int_0^{\infty} \frac{\lambda \sin \lambda x}{\lambda^2 + k^2} d\lambda = \frac{\pi}{2} e^{-kx}, x > 0, k > 0$

(b) Expand $f(x) = x$ as a half range sine series in $0 < x < 2$

VIII. (a) Obtain (i) $L\left(\frac{\cos at - \cos bt}{t}\right)$ (ii) $L^{-1}\left(\frac{(s+2)^2}{(s^2+4s+8)^2}\right)$

(b) Solve $y''' + 2y'' - y' - 2y = 0, y(0) = y'(0) = 0, y''(0) = 6$

OR

IX. (a) Prove that $\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$

(b) Find the inverse Laplace transform of (i) $\log \frac{s+1}{s-1}$ (ii) $\frac{e^{-2s}}{s-3}$

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BTS-III-11.18-1707 Reg.No. [ ]

A

B.Tech. Degree III Semester Examination November 2018

CE/CS/EC/EE/IT/SE/ME
AS 15-1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES
(2015 Schemes)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$.

(b) Explain linear system of equation and when this system has unique solution and no solution.

(c) State and prove one of the property of eigen value of a square matrix.

(d) Define subspace of a vector space with example.

(e) Explain linearly independent and dependent set of vectors in a vector space with example.

(f) Express $f(x)=|x|, -\pi < x < \pi$ as a Fourier series.

(g) Express $f(x)=\begin{cases} kx, & 0 \le x \le \frac{\pi}{2} \\ k(\pi - x), & \frac{\pi}{2} \le x \le \pi \end{cases}$ as half-range sine series.

(h) Find $L\left[t\, e^{-t} \sin t\right]$.

(i) Prove that $L\left[t^n\right] = \dfrac{\lceil n+1}{s^{n+1}}$.

(j) Find $L^{-1}\left[\dfrac{1}{s(s+1)(s+2)}\right]$.

PART B

(4 × 10 = 40)

II. (a) Find the eigen values and eigen vectors of $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$. (5)

(b) Diagonalize the matrix $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$. (5)

OR

(P.T.O.)

III. (a) Verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. (5)

(b) Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to the canonical form. (5)

IV. (a) Show that the set $M = M_{m \times n}$ of all $m \times n$ matrix form a vector space with usual addition and scalar multiplication. (6)

(b) Find a basis and dimension of $M_{2\times2}$. (4)

OR

V. (a) Consider the following polynomials in p(t) with inner product (5)

$$< f,\ g > = \int_0^1 f(t)g(t)dt,\ f(t) = t+2,\ \ g(t) = 3t-2,\ h(t) = t^2 - 2t - 3$$

Find $< f,\ g >$ and $< f,\ h >$.

(b) Let S consists of the following vectors in R^4, (5)

$u_1 = (1, 1, 0, 1),\ u_2 = (1, 2, 1, 3),\ \ u_3 = (1, 1, -9, 2),\ u_4 = (16, -13, 1, 3)$

Show that S is orthogonal and a basis of R^4.

VI. (a) If $f(x)$ is a periodic function defined over a period $(0, 2\pi)$ by (6)

$f(x) = \dfrac{1}{12}\left(3x^2 - 6x\pi + 2\pi^2\right)$. Prove that $f(x) = \sum_{n=1}^{\infty} \dfrac{\cos nx}{n^2}$ and hence show that

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\ldots\ldots\ldots..$$

(b) Find Fourier cosine and sine transform of $f(x) = e^{-ax}$. (4)

OR

VII. (a) Find the Fourier series for $f(x)$, (5)

given $f(x) = \begin{cases} 0 \text{ in } -1 < x < 0 \\ 1 \text{ in } 0 < x < 1 \end{cases}$ and $f(x+2) = f(x)$ for all x.

(b) Find the Fourier integral of $f(x) = \begin{cases} 1 \text{ for } |x| \le 1 \\ 0 \text{ for } |x| > 1 \end{cases}$ (5)

Hence evaluate $\displaystyle\int_0^{\infty} \frac{\sin\lambda \cos\lambda x d\lambda}{\lambda}$.

VIII. (a) Using Laplace transform solve

$y'' + 2y' - 3y = \sin t,\ y = 0,\ y'(0) = 0$ when $t = 0$. (5)

(b) $\dfrac{dy}{dt} + 2y + \int_0^t y\ dt = 2\cos t,\ y(0) = 1.$ (5)

OR

IX. (a) Prove that $\beta(m,n) = \dfrac{\Gamma m\ \Gamma n}{\Gamma m+n}$. (4)

(b) Find the inverse Laplace transform of (6)

(i) $\dfrac{s\ e^{-s/2} + \pi e^{-s}}{s^2 + \pi^2}$ (ii) $\dfrac{e^{-cs}}{s^2(s+a)}$.

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BTS–III(S)–04.18–0782 Reg. No. | | | | | | | | |

B

B.Tech. Degree III Semester Supplementary Examination April 2018

CE/CS/EC/EE/IT/ME/SE

AS 15 -1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES

(*2015 Scheme*)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Prove that the eigen values of an idempotent matrix are either zero or unity.

(b) Determine the values of λ for which the following set of equations may possess non trival solution $3x_1 + x_2 - \lambda x_3 = 0$, $4x_1 - 2x_2 - 3x_3 = 0$, $2\lambda x_1 + 4x_2 + \lambda x_3 = 0$.

(c) Find a non zero vector **w** that is orthogonal to $u = (1,2,1)$ and v = (2,5,4) in R^3.

(d) V=R^n and W=$\{(x_1, x_2 ......x_n) | x_1 x_2 = 0\}$. Is W a subspace of V?

(e) Define half range sine and cosine series.

(f) Using Fourier sine integral show that

$$\int_0^\infty \frac{1-\cos \pi\omega}{\omega} \sin \omega x \, d\omega = \begin{cases} \frac{\pi}{2}, 0 < x < \pi \\ 0, x > \pi \end{cases}.$$

(g) Express $f(x) = \frac{x}{2}$ as a Fourier series in the interval $-\pi < x < \pi$.

(h) Find inverse Laplace transform of $\cot^{-1}(s/2)$.

(i) Find Laplace transform of $\frac{1-e^t}{t}$.

(j) Define unit step function and find Laplace transform of unit step function.

PART B

$(4 \times 10 = 40)$

II. (a) Test for consistency and solve $2x + y - z = 0$, $2x + 5y + 7z = 52$, $x + y + z = 9$.

(b) Find eigen values and eigen vectors of $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.

OR

III. (a) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and hence find A^{-1}.

(b) Diagonalize $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ and find the modal matrix.

(P.T.O.)

IV. (a) Define linear transformation and check whether $T:R^2 \to R$ defined by $T(x,y) = xy$ is linear.

(b) Let V be the vector space of polynomials f(t) with inner product $< f, g > = \int_{-1}^{1} f(t)g(t)dt$. Apply the Gram Schmidt orthogonalization process to $\{1, t, t^2, t^3\}$ to find an orthogonal basis.

OR

V. (a) Define basis and dimension. Also find the value of λ for which the set of vectors $(1,\lambda,5),(1,0,1)$ and $(2,0,1)$ of R^3 form a basis.

(b) Show that (1,1,2), (1,2,5), (5,3,4) are linearly dependent and also express the relation connecting them.

VI. (a) Find a Fourier series to represent $x\text{-}x^2$ from $x = -\pi$ to $x = \pi$. Hence show that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + ... = \pi^2/12.$

(b) Find the Fourier Sine transform of $e^{-|x|}$. Hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}, m > 0$.

OR

VII. (a) Obtain the Fourier cosine series of $f(x) = \begin{cases} x & when\ 0 < x < \frac{\pi}{2} \\ 0 & when\ \frac{\pi}{2} < x < \pi \end{cases}$.

(b) Find the Fourier series expansion of $f(x) = \frac{\pi - x}{2}$ in $0 < x < 2$.

VIII. (a) Define gamma function. Also find $\Gamma(1/2)$.

(b) Find Laplace transform of (i) t sin3t cos2t (ii) $\int_0^\infty t^3 e^{-t} \sin t \, dt$.

OR

IX. (a) Solve $y'' + 5y' + 6y = 5e^t, y(0) = 2, y'(0) = 1$.

(b) Apply convolution theorem to evaluate $L^{-1}\left\{\frac{1}{(s^2+1)(s^2+9)}\right\}$.

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BTS-III(15)-11.17-1554 Reg. No. [| | | | | | |]

B

B.Tech. Degree III Semester Examination November 2017

CE/CS/EC/EE/IT/ME/SE
AS 15-1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Define rank of a matrix and hence find the rank of $\begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 3 & -3 & -4 \\ 5 & 3 & 3 & 11 \end{bmatrix}$.

(b) Find the eigen values and eigen vector of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

(c) Check whether the vectors $(1, 3, 2)$, $(5, -2, 1)$, $(-7, 13, 4)$ are linearly dependent if so find the relation connecting them.

(d) Verify whether the collection of all polynomials of degree n form a vector space or not under usual addition and sealer multiplication of Polynomials. Justify.

(e) Show that the Fourier series for $f(x) = x, -\pi < x < \pi$ is given by $f(x) = 2\sum_{n=1}^{\infty} \frac{(-1)^{n+1}\sin nx}{n}$

(f) Find the Fourier cosine transform of $5e^{-2x} + 2e^{-5x}$.

(g) Obtain the half range sine series of the function $f(x) = kx(x-l)$ in $0 \le x \le l$.

(h) Find the Laplace transform of $t\cos^3 t$.

(i) Find the inverse Laplace transform of $\log\left[\frac{s^2+a^2}{s^2-b^2}\right]$.

(j) Using convolution theorem, find the inverse Laplace transform of $\frac{s}{(s^2+a^2)^2}$.

PART B

(4 × 10 = 40)

II. (a) Test for consistency and solve $4x - 2y + 6z = 8$; $x + y - 3z = -1$; $15x - 3y + 9z = 21$.

(b) Diagonalize the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$.

OR

(P.T.O.)

III. (a) Using Cayley-Hamilton theorem find the inverse of $\begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$.

(b) Reduce the quadratic form $3x_1^2+3x_2^2+3x_3^2+2x_1x_2+2x_1x_3-2x_2x_3$ to Canonical form.

IV. (a) Suppose the vectors u, v, w are linearly independent. Show that the vectors $u+v,\ u-v,\ u-2v+w$ are also linearly independent.

(b) Let V be the vector space of 2×2 matrices. Let W be the subspace of symmetric matrices. Show that dim W = 3, by finding a basis of W.

OR

V. (a) Suppose the mapping $F: R^2 \to R^2$ is defined by $F(x, y) = (x+y, x)$. Show that F is a linear transformation.

(b) Let S be a subset of an inner product space V. Then prove that the orthogonal complement of S is a subspace of V.

VI. (a) Find the Fourier series for $f(x) = x^2$ in $-1 < x < 1$.

(b) Express the function $f(x) = \begin{cases} 1 & \text{for } |x| \le | \\ 0 & \text{for } |x| > | \end{cases}$ as a Fourier integral. Hence evaluate $\int_0^\infty \frac{\sin\lambda \cos\lambda x \, d\lambda}{\lambda}$.

OR

VII. (a) Find the Fourier sine transform of e^{-x}, $x \ge 0$. Hence evaluate $\int_0^\infty \frac{x \sin mx}{1+x^2} dx$.

(b) Show the Fourier transform of $f(x) = e^{\frac{-x^2}{2}}$ is $e\frac{-s^2}{2}$.

VIII. (a) Solve (using Laplace transform)
$y'' + y' = t^2 + 2t,\ y(0) = 4,\ y'(0) = -2$.

(b) Find the Laplace transform of periodic function.

OR

IX. (a) Solve $\frac{dy}{dt} + 3y + 2\int_0^t y dt = t$ for which $y(0) = 0$.

(b) Using Laplace transform evaluate $\int_0^\infty \frac{e^{-2t} - e^{-3t}}{t} dt$.

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BTS-III(S)-05.17-0718 Reg. No. | | | | | | | | |

C

B.Tech. Degree III Semester Supplementary Examination May 2017

IT/CS/EC/CE/EE/ME/SE

AS 15-1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Find the values of l and m such that the rank of the matrix $\begin{bmatrix} 2 & 1 & -1 & 3 \\ 1 & -1 & 2 & 4 \\ 7 & -1 & l & m \end{bmatrix}$ is 2.

(b) Find the value of λ for which the system of equation $x+2y=0;\ 2x+\lambda y=0$ has (i) unique solution and (ii) more than one solution.

(c) Find the eigen values and eigen vectors of $A=\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

(d) Explain linearly independent and dependent vectors with examples.

(e) Define basis and dimension of a vector space with example.

(f) Find the Fourier series to represent $x-\pi$ in the interval $(-\pi, \pi)$.

(g) Obtain the half range sine series of the function $f(x)=kx(x-l)$ in $0 \le x \le l$.

(h) Find the Laplace transform of $t\sin 2t$.

(i) Find the Laplace transform of $\frac{1-e^t}{t}$.

(j) Find the inverse Laplace transform of $\log\left[\frac{1+s}{s^2}\right]$.

PART B

(4 × 10 = 40)

II. (a) Diagonalize the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

(b) Reduce the quadratic form $2xy+2yz+2zx$ into canonical form.

OR

III. (a) Test for consistency and solve the following;
$2xy-y-z=2;\ x+2y+z=2;\ 4x-7y-5z=2$.

(b) Verify Cayley Hamilton theorem for the matrix $A=\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$.

(P.T.O.)

IV. (a) Explain inner product and inner product space with examples.

(b) Explain orthogonal and orthonormal basis in an inner product space with examples.

OR

V. (a) Explain Gram Schimidt orthogonalization process.

(b) Find k so that $u=(1, 2, k, 3)$ and $v=(3, k, 7, -5)$ in R^4 are orthogonal.

VI. (a) Find the Fourier series expansion of period $2l$ for the function $f(x)=(l-x)^2$ in the range $(0, 2l)$.

(b) Using Fourier sine integral for $f(x)=e^{-ax}(a>0)$ show that $\int_0^\alpha \frac{\lambda \sin \lambda x}{\lambda^2+a^2} d\lambda = \frac{\pi}{2} e^{-ax}$.

OR

VII. (a) Find the Fourier cosine integral of the function e^{-ax} and hence deduce the value of the integral $\int_0^\alpha \frac{\cos \lambda x}{1+\lambda^2} d\lambda$.

(b) Find the Fourier transform of $f(x)=\begin{cases} 1 & \text{in } |x|<a \\ 0 & \text{in } |x|>a \end{cases}$.

VIII. (a) Using convolution theorem find $L^{-1}\left[\frac{1}{S(S^2+1)}\right]$.

(b) Using Laplace transfer solve $y''-3y'+2y=e^{2t}$, $y(0)=-3$, $y'(0)=5$.

OR

IX. (a) Using Laplace transform solve the integral equation $\frac{dy}{dt}+4y+5\int_0^t y\,dt=e^{-t}$, when $y(0)=0$.

(b) Use Laplace transform to evaluate $\int_0^\infty t\, e^{-2t} \sin t\, dt$.

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BTS – III(15)–11.16-1508 Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ]

A

B.Tech. Degree III Semester Examination November 2016

CE/CS/EC/EE/IT/ME/SE
AS 15–1301 LINEAR ALGEBRA AND TRANSFORM TECHNIQUES
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

$10 \times 2 = 20$

I. (a) Prove that every square matrix and its transpose have same eigen values.

(b) Explain when a linear system of equations has unique solution, no solution and more than one solution.

(c) If x, y, z are linearly independent vectors of a vector space V prove that $x,\ x+y,\ x+y+z$ are also linearly independent.

(d) Define basis and dimension of a vector space with example.

(e) Express $f(x) = x^2, -\pi < x < \pi$ as a Fourier series.

(f) Find the Fourier transform of $f(x) = \begin{cases} 1 & if \quad |x| < a \\ 0 & if \quad |x| > a \end{cases}$

(g) Find the half range cosine series for $f(x) = x.$ is $(0, \pi)$ and deduce the sum $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots.$

(h) Find the Laplace transform of tSin3tCos2t.

(i) Find the Laplace transform of $\log\left[\frac{S(S+1)}{S^2+1}\right]$.

(j) Find the Laplace transform of periodic function.

PART B

$4 \times 10 = 40)$

II. (a) State Cayley-Hamilton theorem and verify it for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1}.

(b) Diagonalize the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ hence find A^4.

OR

(P.T.O.)

III. (a) Reduce the quadratic form $3x^2+5y^2+3z^2-2yz+2zx-2xy$ to a Canonical form by orthogonal reduction.

(b) Find the value of λ and μ for which the equations $2x+3y+5z=9$; $7x+3y-2z=8$; $2x+3y+\lambda z=\mu$ have (i) no solution (ii) unique solution (iii) more than one solution.

IV. (a) Define vector space, subspace and give an example of a vector space having a subset which is a subspace and a subset which is not a subspace. Justify your answer.

(b) Let W be any plane is R^3 passing through the origin. Verify whether W is a subspace of R^3.

OR

V. (a) Apply Gram-Schmidt orthogonalization process to find an orthogonal basis and an orthonormal basis for the subspace U of R^4 spanned by $V_1=(1,1,1,1)$, $V_2=(1,2,4,5)$, $V_3=(1,{}^-3,{}^-4,{}^-2)$

(b) Let T be a map from R^3 to R defined by $T(x,y,z)=x^2+y^2+z^2$. Verify whether T is a linear transformation.

VI (a) Find the Fourier series for $f(x)$, given $f(x)=\begin{cases}0 \; in -1<x<0\\ 1 \; in \; 0<x<1\end{cases}$ and $f(x+2)=f(x)$ for all x.

(b) Using Fourier integral for $f(x)=e^{-ax}(a>0)$ show that $\int_o^\infty \frac{\lambda \; Sin \lambda x}{\lambda^2+a^2} d\lambda = \frac{\pi}{2}e^{-ax}$.

OR

VII. (a) Find the Fourier transform of $f(x)=\begin{cases}1-x^2 \; in|x|\le l\\ 0 \quad in|x|>l\end{cases}$. Hence prove that

$$\int_o^\infty \frac{SinS-S\cos S}{S^3}\cos\frac{s}{2}\; ds=\frac{3\pi}{16}.$$

(b) Find the half range sine series for the function $f(x)=\begin{cases}x-1 & 0\le x\le 1\\ 1-x, & 1\le x\le 2\end{cases}$

VIII. (a) State convolution theorem and use it to find the inverse Laplace transform of $\frac{1}{(S^2+4)^2}$.

(b) Solve (using transform) $y''+2y'-3y=\sin t$, *given* $y=0,\ y'(0)=0$ *when* $t=0$.

OR

IX. (a) Determine y which satisfies the equation $\frac{dx}{dt}+2y+\int_o^t y\; dt=2\cos t,\ y(0)=1$.

(b) Prove that $\beta(m,n)=\frac{\Gamma m \, \Gamma n}{\Gamma(m+n)}$.

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